

Análisis Exploratorio, Limpieza y Transformación de Datos (KDD)

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1. Introducción

Este informe presenta un análisis de datos basado en la metodología KDD aplicado a un dataset de jugadores del mundial 2022, construido a partir de información de SOFIFA.

El objetivo es limpiar, explorar, analizar y transformar los datos para obtener conocimiento relevante sobre el rendimiento de los jugadores.

2. Carga y Exploración de Datos

```
data <- read_csv("FC26_20250921.csv")
```

```
## Rows: 751 Columns: 129
## -- Column specification -----
## Delimiter: ","
## chr  (54): short_name_x, long_name, player_positions_x, nationality_name_x, ...
## dbl  (73): sofifa_id_x, overall_x, potential_x, age_x, nation_jersey_number...
## date  (2): dob, club_joined
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
dim(data)
```

```
## [1] 751 129
```

```
glimpse(data)
```

```
## Rows: 751
## Columns: 129
## $ sofifa_id_x      <dbl> 237942, 175254, 223053, 189454, 191164, 22~
## $ short_name_x     <chr> "P. Estupiñán", "F. Caicedo", "Á. Mena", "~
## $ long_name        <chr> "Pervis Josué Estupiñán Tenorio", "Felipe ~
## $ player_positions_x <chr> "LB, LWB", "CF, ST", "RM, RW", "CAM", "ST,~
```

```

## $ overall_x      <dbl> 79, 78, 78, 76, 75, 75, 74, 74, 73, 72, 72~
## $ potential_x    <dbl> 85, 78, 78, 76, 75, 75, 74, 84, 80, 73, 75~
## $ age_x          <dbl> 23, 32, 33, 35, 31, 31, 30, 20, 22, 27, 26~
## $ nationality_name_x <chr> "Ecuador", "Ecuador", "Ecuador", "Ecuador"~
## $ nation_position_x <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ nation_jersey_number_x <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ club_name_x     <chr> "Villarreal CF", "Genoa", "Club León", "Ba~
## $ league_name_x   <chr> "Spain Primera Division", "Italian Serie A~
## $ value_eur_x     <dbl> 26000000, 11500000, 9000000, 3100000, 4800~
## $ wage_eur_x      <dbl> 29000, 22000, 28000, 1000, 23000, 45000, 3~
## $ pace_x          <dbl> 82, 74, 81, 74, 88, 88, 87, 85, 82, 72, 61~
## $ shooting_x      <dbl> 34, 78, 74, 65, 72, 74, 64, 66, 48, 68, 46~
## $ passing_x       <dbl> 68, 70, 72, 76, 65, 66, 68, 65, 62, 72, 61~
## $ dribbling_x     <dbl> 72, 75, 82, 77, 74, 73, 75, 79, 62, 72, 67~
## $ defending_x      <dbl> 75, 44, 45, 44, 32, 46, 53, 43, 68, 40, 68~
## $ physic_x        <dbl> 75, 80, 49, 68, 69, 75, 73, 64, 66, 47, 73~
## $ sofifa_id_y     <dbl> 237942, 175254, 223053, 189454, 191164, 22~
## $ player_url      <chr> "https://sofifa.com/player/237942/pervis-e~
## $ short_name_y    <chr> "P. Estupiñán", "F. Caicedo", "Á. Mena", "~
## $ player_positions_y <chr> "LB, LWB", "CF, ST", "RM, RW", "CAM", "ST,~
## $ overall_y       <dbl> 79, 78, 78, 76, 75, 75, 74, 74, 73, 72, 72~
## $ potential_y     <dbl> 85, 78, 78, 76, 75, 75, 74, 84, 80, 73, 75~
## $ value_eur_y     <dbl> 26000000, 11500000, 9000000, 3100000, 4800~
## $ wage_eur_y      <dbl> 29000, 22000, 28000, 1000, 23000, 45000, 3~
## $ age_y           <dbl> 23, 32, 33, 35, 31, 31, 30, 20, 22, 27, 26~
## $ dob             <date> 1998-01-21, 1988-09-05, 1988-01-21, 1986--
## $ height_cm       <dbl> 175, 183, 168, 174, 179, 177, 175, 178, 17~
## $ weight_kg       <dbl> 78, 82, 71, 67, 74, 74, 76, 65, 60, 66, 71~
## $ club_team_id    <dbl> 483, 110556, 110781, 110981, 111678, 326, ~
## $ club_name_y     <chr> "Villarreal CF", "Genoa", "Club León", "Ba~
## $ league_name_y   <chr> "Spain Primera Division", "Italian Serie A~
## $ league_level    <dbl> 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, ~
## $ club_position   <chr> "SUB", "SUB", "RM", "CAM", "LS", "ST", "SU~
## $ club_jersey_number <dbl> 12, 9, 13, 10, 10, 13, 30, 12, 26, 11, 8, ~
## $ club_loaned_from <chr> NA, NA, NA, NA, NA, NA, NA, NA, "Sporting CP",~
## $ club_joined     <date> 2020-09-16, 2021-09-01, 2019-01-01, 2016--
## $ club_contract_valid_until <dbl> 2027, 2026, 2021, 2021, 2022, 2023, 2025, ~
## $ nationality_id   <dbl> 57, 57, 57, 57, 57, 57, 57, 57, 57, 57, 57~
## $ nationality_name_y <chr> "Ecuador", "Ecuador", "Ecuador", "Ecuador"~
## $ nation_team_id  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ nation_position_y <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ nation_jersey_number_y <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ preferred_foot  <chr> "Left", "Left", "Left", "Right", "Left", "~
## $ weak_foot       <dbl> 3, 3, 4, 4, 3, 3, 3, 3, 2, 3, 3, 4, 3, 4, ~
## $ skill_moves     <dbl> 3, 3, 4, 4, 4, 4, 3, 4, 3, 3, 2, 3, 3, 3, ~
## $ international_reputation <dbl> 1, 2, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ work_rate       <chr> "High/Medium", "High/Medium", "High/Medium~
## $ body_type       <chr> "Stocky (170-185)", "Stocky (170-185)", "L~
## $ real_face       <chr> "No", "No", "No", "No", "No", "No", "No", "Yes",~
## $ release_clause_eur <dbl> 59800000, 20700000, 15300000, 6800000, 820~
## $ player_tags     <chr> NA, "#Strength", "#Acrobat", NA, NA, NA, N~
## $ player_traits   <chr> "Speed Dribbler (AI)", "Flair", "Finesse S~
## $ pace_y          <dbl> 82, 74, 81, 74, 88, 88, 87, 85, 82, 72, 61~
## $ shooting_y      <dbl> 34, 78, 74, 65, 72, 74, 64, 66, 48, 68, 46~

```

```

## $ passing_y          <dbl> 68, 70, 72, 76, 65, 66, 68, 65, 62, 72, 61~
## $ dribbling_y        <dbl> 72, 75, 82, 77, 74, 73, 75, 79, 62, 72, 67~
## $ defending_y         <dbl> 75, 44, 45, 44, 32, 46, 53, 43, 68, 40, 68~
## $ physic_y           <dbl> 75, 80, 49, 68, 69, 75, 73, 64, 66, 47, 73~
## $ attacking_crossing  <dbl> 79, 59, 75, 69, 66, 67, 72, 68, 70, 68, 43~
## $ attacking_finishing <dbl> 20, 79, 77, 73, 76, 75, 60, 65, 50, 68, 41~
## $ attacking_heading_accuracy <dbl> 75, 77, 69, 48, 67, 74, 47, 41, 67, 43, 54~
## $ attacking_short_passing <dbl> 78, 79, 75, 78, 70, 68, 72, 68, 67, 73, 72~
## $ attacking_volleys   <dbl> 38, 78, 71, 60, 72, 62, 60, 71, 35, 59, 37~
## $ skill_dribbling     <dbl> 73, 77, 82, 80, 75, 72, 76, 79, 53, 71, 60~
## $ skill_curve         <dbl> 59, 66, 64, 62, 60, 73, 63, 76, 41, 70, 48~
## $ skill_fk_accuracy   <dbl> 36, 68, 67, 75, 62, 67, 61, 52, 35, 71, 47~
## $ skill_long_passing  <dbl> 62, 66, 71, 77, 58, 60, 62, 46, 55, 71, 69~
## $ skill_ball_control  <dbl> 71, 79, 80, 76, 76, 72, 75, 77, 69, 73, 70~
## $ movement_acceleration <dbl> 81, 73, 81, 72, 84, 89, 85, 86, 80, 73, 68~
## $ movement_sprint_speed <dbl> 83, 74, 81, 75, 92, 88, 88, 85, 83, 71, 55~
## $ movement_agility   <dbl> 76, 59, 91, 77, 80, 84, 80, 88, 74, 70, 78~
## $ movement_reactions  <dbl> 71, 79, 77, 70, 69, 73, 65, 73, 65, 66, 69~
## $ movement_balance   <dbl> 70, 67, 85, 72, 62, 72, 72, 80, 89, 94, 85~
## $ power_shot_power    <dbl> 41, 84, 69, 57, 65, 80, 70, 70, 50, 67, 61~
## $ power_jumping       <dbl> 69, 75, 33, 57, 78, 91, 71, 51, 66, 63, 69~
## $ power_stamina       <dbl> 79, 59, 75, 78, 75, 81, 82, 69, 90, 70, 73~
## $ power_strength      <dbl> 74, 91, 47, 69, 70, 76, 70, 63, 59, 37, 69~
## $ power_long_shots    <dbl> 42, 70, 72, 52, 64, 70, 70, 65, 42, 68, 42~
## $ mentality_aggression <dbl> 74, 80, 27, 58, 56, 61, 71, 61, 52, 40, 86~
## $ mentality_interceptions <dbl> 75, 43, 43, 42, 45, 40, 56, 40, 67, 26, 69~
## $ mentality_positioning <dbl> 52, 82, 78, 71, 79, 74, 65, 73, 59, 71, 55~
## $ mentality_vision    <dbl> 56, 68, 69, 84, 63, 65, 65, 71, 60, 78, 58~
## $ mentality_penalties <dbl> 69, 73, 72, 77, 82, 68, 54, 60, 40, 73, 49~
## $ mentality_composure <dbl> 62, 77, 75, 75, 66, 63, 63, 78, 55, 61, 65~
## $ defending_marking_awareness <dbl> 69, 32, 38, 57, 20, 50, 57, 43, 69, 62, 68~
## $ defending_standing_tackle <dbl> 80, 44, 46, 34, 25, 40, 52, 46, 70, 29, 71~
## $ defending_sliding_tackle <dbl> 79, 47, 41, 33, 30, 38, 46, 38, 64, 30, 73~
## $ goalkeeping_diving  <dbl> 14, 7, 6, 11, 7, 9, 6, 13, 6, 14, 9, 5, 13~
## $ goalkeeping_handling <dbl> 7, 13, 10, 15, 12, 12, 9, 8, 12, 11, 10, 1~
## $ goalkeeping_kicking <dbl> 14, 7, 9, 14, 6, 16, 14, 14, 9, 7, 12, 9, ~
## $ goalkeeping_positioning <dbl> 7, 8, 10, 11, 16, 9, 13, 10, 10, 8, 12, 7, ~
## $ goalkeeping_reflexes <dbl> 7, 10, 13, 7, 15, 8, 8, 12, 6, 12, 11, 9, ~
## $ goalkeeping_speed   <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ ls                  <chr> "57+2", "79-1", "75+2", "68+2", "74+1", "7~
## $ st                  <chr> "57+2", "79-1", "75+2", "68+2", "74+1", "7~
## $ rs                  <chr> "57+2", "79-1", "75+2", "68+2", "74+1", "7~
## $ lw                  <chr> "65", "75", "78", "74", "75", "74", "72", ~
## $ lf                  <chr> "61", "78", "77", "73", "74", "74", "70", ~
## $ cf                  <chr> "61", "78", "77", "73", "74", "74", "70", ~
## $ rf                  <chr> "61", "78", "77", "73", "74", "74", "70", ~
## $ rw                  <chr> "65", "75", "78", "74", "75", "74", "72", ~
## $ lam                 <chr> "64+2", "76+2", "77+1", "76", "72+2", "72+~
## $ cam                 <chr> "64+2", "76+2", "77+1", "76", "72+2", "72+~
## $ ram                 <chr> "64+2", "76+2", "77+1", "76", "72+2", "72+~
## $ lm                  <chr> "69+2", "74+2", "77+1", "75+1", "74+1", "7~
## $ lcm                 <chr> "67+2", "71+2", "72+2", "72+2", "66+2", "6~
## $ cm                  <chr> "67+2", "71+2", "72+2", "72+2", "66+2", "6~
## $ rcm                 <chr> "67+2", "71+2", "72+2", "72+2", "66+2", "6~

```

```
## $ rm <chr> "69+2", "74+2", "77+1", "75+1", "74+1", "7~
## $ lwb <chr> "77+2", "60+2", "64+2", "62+2", "58+2", "6~
## $ ldm <chr> "73+2", "62+2", "58+2", "62+2", "53+2", "5~
## $ cdm <chr> "73+2", "62+2", "58+2", "62+2", "53+2", "5~
## $ rdm <chr> "73+2", "62+2", "58+2", "62+2", "53+2", "5~
## $ rwb <chr> "77+2", "60+2", "64+2", "62+2", "58+2", "6~
## $ lb <chr> "77+2", "58+2", "61+2", "58+2", "55+2", "6~
## $ lcb <chr> "75+2", "59+2", "50+2", "52+2", "48+2", "5~
## $ cb <chr> "75+2", "59+2", "50+2", "52+2", "48+2", "5~
## $ rcb <chr> "75+2", "59+2", "50+2", "52+2", "48+2", "5~
## $ rb <chr> "77+2", "58+2", "61+2", "58+2", "55+2", "6~
## $ gk <chr> "16+2", "17+2", "17+2", "18+2", "18+2", "1~
## $ player_face_url <chr> "https://cdn.sofifa.net/players/237/942/22~
## $ club_logo_url <chr> "https://cdn.sofifa.net/teams/483/60.png",~
## $ club_flag_url <chr> "https://cdn.sofifa.net/flags/es.png", "ht~
## $ nation_logo_url <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
## $ nation_flag_url <chr> "https://cdn.sofifa.net/flags/ec.png", "ht~
```

```
summary(data)
```

```
##      sofifa_id_x      short_name_x      long_name      player_positions_x
## Min.   : 20801      Length:751      Length:751      Length:751
## 1st Qu.:201216      Class :character      Class :character      Class :character
## Median :218540      Mode  :character      Mode  :character      Mode  :character
## Mean   :216031
## 3rd Qu.:233212
## Max.   :264640
##
##      overall_x      potential_x      age_x      nationality_name_x
## Min.   :62.00      Min.   :62.00      Min.   :17.00      Length:751
## 1st Qu.:72.00      1st Qu.:74.00      1st Qu.:25.00      Class :character
## Median :76.00      Median :79.00      Median :27.00      Mode  :character
## Mean   :76.27      Mean   :78.63      Mean   :27.19
## 3rd Qu.:80.00      3rd Qu.:83.00      3rd Qu.:30.00
## Max.   :93.00      Max.   :95.00      Max.   :38.00
##
##      nation_position_x      nation_jersey_number_x      club_name_x
## Length:751      Min.   : 1.00      Length:751
## Class :character      1st Qu.: 7.00      Class :character
## Mode  :character      Median :12.00      Mode  :character
## Mean   :12.45
## 3rd Qu.:18.00
## Max.   :28.00
## NA's   :382
##      league_name_x      value_eur_x      wage_eur_x      pace_x
## Length:751      Min.   : 160000      Min.   : 500      Min.   :32.00
## Class :character      1st Qu.: 2700000      1st Qu.: 12000      1st Qu.:66.00
## Mode  :character      Median : 8000000      Median : 26000      Median :74.00
## Mean   :17694744      Mean   : 45770      Mean   :73.48
## 3rd Qu.:24000000      3rd Qu.: 57000      3rd Qu.:81.00
## Max.   :194000000      Max.   :350000      Max.   :97.00
## NA's   :10      NA's   :10      NA's   :77
##      shooting_x      passing_x      dribbling_x      defending_x      physic_x
## Min.   :22.00      Min.   :38.00      Min.   :40.0      Min.   :20.00      Min.   :40.00
```

```

## 1st Qu.:54.00 1st Qu.:62.00 1st Qu.:68.0 1st Qu.:43.00 1st Qu.:67.00
## Median :67.00 Median :68.00 Median :73.0 Median :65.00 Median :73.00
## Mean :62.84 Mean :67.94 Mean :72.6 Mean :59.11 Mean :71.61
## 3rd Qu.:73.00 3rd Qu.:75.00 3rd Qu.:79.0 3rd Qu.:74.00 3rd Qu.:78.00
## Max. :94.00 Max. :93.00 Max. :95.0 Max. :91.00 Max. :89.00
## NA's :77 NA's :77 NA's :77 NA's :77 NA's :77
## sofifa_id_y player_url short_name_y player_positions_y
## Min. : 20801 Length:751 Length:751 Length:751
## 1st Qu.:201181 Class :character Class :character Class :character
## Median :218540 Mode :character Mode :character Mode :character
## Mean :216013
## 3rd Qu.:233212
## Max. :264640
##
## overall_y potential_y value_eur_y wage_eur_y
## Min. :62.00 Min. :62.00 Min. : 160000 Min. : 500
## 1st Qu.:72.00 1st Qu.:74.00 1st Qu.: 2700000 1st Qu.: 11000
## Median :76.00 Median :79.00 Median : 8000000 Median : 26000
## Mean :76.26 Mean :78.62 Mean : 17692314 Mean : 45705
## 3rd Qu.:80.00 3rd Qu.:83.00 3rd Qu.: 24000000 3rd Qu.: 57000
## Max. :93.00 Max. :95.00 Max. :194000000 Max. :350000
## NA's :10 NA's :10
## age_y dob height_cm weight_kg
## Min. :17.00 Min. :1983-02-08 Min. :165.0 Min. : 56.00
## 1st Qu.:25.00 1st Qu.:1991-04-11 1st Qu.:177.0 1st Qu.: 70.00
## Median :27.00 Median :1993-11-11 Median :182.0 Median : 76.00
## Mean :27.19 Mean :1993-10-27 Mean :181.9 Mean : 76.26
## 3rd Qu.:30.00 3rd Qu.:1996-06-21 3rd Qu.:187.0 3rd Qu.: 81.00
## Max. :38.00 Max. :2003-10-16 Max. :200.0 Max. :100.00
##
## club_team_id club_name_y league_name_y league_level
## Min. : 1 Length:751 Length:751 Min. :1.000
## 1st Qu.: 73 Class :character Class :character 1st Qu.:1.000
## Median : 452 Mode :character Mode :character Median :1.000
## Mean : 21329 Mean :1.073
## 3rd Qu.: 1860 3rd Qu.:1.000
## Max. :114162 Max. :4.000
## NA's :10 NA's :10
## club_position club_jersey_number club_loaned_from club_joined
## Length:751 Min. : 1.0 Length:751 Min. :2005-01-01
## Class :character 1st Qu.: 8.0 Class :character 1st Qu.:2017-08-15
## Mode :character Median :14.0 Mode :character Median :2019-07-20
## Mean :17.5 Mean :2019-01-03
## 3rd Qu.:23.0 3rd Qu.:2020-10-16
## Max. :99.0 Max. :2021-09-14
## NA's :10 NA's :42
## club_contract_valid_until nationality_id nationality_name_y nation_team_id
## Min. :2021 Min. : 7.00 Length:751 Min. : 1318
## 1st Qu.:2022 1st Qu.: 37.50 Class :character 1st Qu.: 1336
## Median :2023 Median : 57.00 Mode :character Median : 1364
## Mean :2023 Mean : 79.38 Mean : 14719
## 3rd Qu.:2024 3rd Qu.:129.00 3rd Qu.: 1386
## Max. :2027 Max. :195.00 Max. :111455
## NA's :10 NA's :383

```

```

## nation_position_y nation_jersey_number_y preferred_foot weak_foot
## Length:751 Min. : 1.00 Length:751 Min. :1.000
## Class :character 1st Qu.: 6.75 Class :character 1st Qu.:3.000
## Mode :character Median :12.00 Mode :character Median :3.000
## Mean :12.45 Mean :3.229
## 3rd Qu.:18.00 3rd Qu.:4.000
## Max. :28.00 Max. :5.000
## NA's :383
## skill_moves international_reputation work_rate body_type
## Min. :1.000 Min. :1.00 Length:751 Length:751
## 1st Qu.:2.000 1st Qu.:1.00 Class :character Class :character
## Median :3.000 Median :1.00 Mode :character Mode :character
## Mean :2.828 Mean :1.68
## 3rd Qu.:3.000 3rd Qu.:2.00
## Max. :5.000 Max. :5.00
##
## real_face release_clause_eur player_tags player_traits
## Length:751 Min. : 263000 Length:751 Length:751
## Class :character 1st Qu.: 4400000 Class :character Class :character
## Mode :character Median : 15800000 Mode :character Mode :character
## Mean : 34206392
## 3rd Qu.: 45100000
## Max. :373500000
## NA's :42
## pace_y shooting_y passing_y dribbling_y
## Min. :32.00 Min. :22.00 Min. :38.00 Min. :40.00
## 1st Qu.:66.00 1st Qu.:54.00 1st Qu.:62.00 1st Qu.:68.00
## Median :74.00 Median :67.00 Median :68.00 Median :73.00
## Mean :73.47 Mean :62.85 Mean :67.93 Mean :72.59
## 3rd Qu.:81.00 3rd Qu.:73.00 3rd Qu.:75.00 3rd Qu.:79.00
## Max. :97.00 Max. :94.00 Max. :93.00 Max. :95.00
## NA's :76 NA's :76 NA's :76 NA's :76
## defending_y physic_y attacking_crossing attacking_finishing
## Min. :20.00 Min. :40.00 Min. : 8.00 Min. : 5.00
## 1st Qu.:43.00 1st Qu.:67.00 1st Qu.:51.00 1st Qu.:41.00
## Median :65.00 Median :73.00 Median :65.00 Median :62.00
## Mean :59.07 Mean :71.61 Mean :58.95 Mean :55.85
## 3rd Qu.:74.00 3rd Qu.:78.00 3rd Qu.:73.00 3rd Qu.:72.50
## Max. :91.00 Max. :89.00 Max. :94.00 Max. :95.00
## NA's :76 NA's :76
## attacking_heading_accuracy attacking_short_passing attacking_volleys
## Min. : 7.00 Min. :11.00 Min. : 5.00
## 1st Qu.:50.00 1st Qu.:66.00 1st Qu.:38.00
## Median :64.00 Median :73.00 Median :58.00
## Mean :59.29 Mean :69.59 Mean :52.77
## 3rd Qu.:74.00 3rd Qu.:78.50 3rd Qu.:69.00
## Max. :92.00 Max. :94.00 Max. :90.00
##
## skill_dribbling skill_curve skill_fk_accuracy skill_long_passing
## Min. : 8.00 Min. : 9.00 Min. : 7.00 Min. :12.00
## 1st Qu.:63.00 1st Qu.:48.00 1st Qu.:37.00 1st Qu.:57.50
## Median :72.00 Median :63.00 Median :54.00 Median :66.00
## Mean :66.11 Mean :58.26 Mean :50.93 Mean :63.82
## 3rd Qu.:78.00 3rd Qu.:73.00 3rd Qu.:67.00 3rd Qu.:74.00

```

```

## Max. :96.00 Max. :93.00 Max. :94.00 Max. :93.00
##
## skill_ball_control movement_acceleration movement_sprint_speed
## Min. : 9.00 Min. :25.00 Min. :18.00
## 1st Qu.:66.00 1st Qu.:63.00 1st Qu.:64.00
## Median :74.00 Median :73.00 Median :73.00
## Mean :69.22 Mean :70.62 Mean :70.75
## 3rd Qu.:79.00 3rd Qu.:81.00 3rd Qu.:81.00
## Max. :96.00 Max. :97.00 Max. :97.00
##
## movement_agility movement_reactions movement_balance power_shot_power
## Min. :29.00 Min. :48.00 Min. :24.00 Min. :24.00
## 1st Qu.:62.00 1st Qu.:68.00 1st Qu.:58.00 1st Qu.:60.00
## Median :72.00 Median :73.00 Median :69.00 Median :71.00
## Mean :70.06 Mean :73.44 Mean :67.34 Mean :68.29
## 3rd Qu.:81.00 3rd Qu.:79.00 3rd Qu.:78.00 3rd Qu.:78.00
## Max. :95.00 Max. :94.00 Max. :95.00 Max. :95.00
##
## power_jumping power_stamina power_strength power_long_shots
## Min. :32.00 Min. :20.00 Min. :31.00 Min. : 6.0
## 1st Qu.:63.00 1st Qu.:67.50 1st Qu.:64.00 1st Qu.:46.0
## Median :72.00 Median :75.00 Median :72.00 Median :64.0
## Mean :70.05 Mean :71.45 Mean :70.78 Mean :57.3
## 3rd Qu.:79.00 3rd Qu.:81.00 3rd Qu.:79.00 3rd Qu.:73.0
## Max. :95.00 Max. :97.00 Max. :95.00 Max. :94.0
##
## mentality_aggression mentality_interceptions mentality_positioning
## Min. :16.00 Min. : 7.0 Min. : 4.00
## 1st Qu.:55.00 1st Qu.:34.5 1st Qu.:53.00
## Median :68.00 Median :63.0 Median :68.00
## Mean :64.02 Mean :54.7 Mean :60.67
## 3rd Qu.:76.00 3rd Qu.:75.0 3rd Qu.:75.00
## Max. :94.00 Max. :91.0 Max. :95.00
##
## mentality_vision mentality_penalties mentality_composure
## Min. :23.00 Min. :10.00 Min. :30.00
## 1st Qu.:58.00 1st Qu.:44.00 1st Qu.:64.00
## Median :67.00 Median :58.00 Median :71.00
## Mean :65.75 Mean :55.36 Mean :70.61
## 3rd Qu.:75.00 3rd Qu.:68.00 3rd Qu.:78.00
## Max. :95.00 Max. :92.00 Max. :96.00
##
## defending_marking_awareness defending_standing_tackle defending_sliding_tackle
## Min. : 6.00 Min. : 9.00 Min. : 8.00
## 1st Qu.:35.00 1st Qu.:34.00 1st Qu.:30.00
## Median :60.00 Median :63.00 Median :60.00
## Mean :54.16 Mean :54.85 Mean :51.48
## 3rd Qu.:73.00 3rd Qu.:75.00 3rd Qu.:72.00
## Max. :92.00 Max. :93.00 Max. :91.00
##
## goalkeeping_diving goalkeeping_handling goalkeeping_kicking
## Min. : 2.00 Min. : 3.0 Min. : 2.00
## 1st Qu.: 8.00 1st Qu.: 9.0 1st Qu.: 8.00
## Median :11.00 Median :11.0 Median :11.00

```

```

## Mean :17.22      Mean :17.1      Mean :16.66
## 3rd Qu.:14.00    3rd Qu.:14.0    3rd Qu.:14.00
## Max. :89.00      Max. :89.0      Max. :91.00
##
## goalkeeping_positioning goalkeeping_reflexes goalkeeping_speed
## Min. : 2.00      Min. : 3.00      Min. :25.00
## 1st Qu.: 8.00      1st Qu.: 8.00      1st Qu.:41.75
## Median :11.00      Median :11.00      Median :47.00
## Mean :17.17      Mean :17.52      Mean :45.92
## 3rd Qu.:14.00      3rd Qu.:14.00      3rd Qu.:51.00
## Max. :89.00      Max. :90.00      Max. :63.00
##
## NA's :675
##
## ls st rs lw
## Length:751 Length:751 Length:751 Length:751
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## lf cf rf rw
## Length:751 Length:751 Length:751 Length:751
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## lam cam ram lm
## Length:751 Length:751 Length:751 Length:751
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## lcm cm rcm rm
## Length:751 Length:751 Length:751 Length:751
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## lbw ldm cdm rdm
## Length:751 Length:751 Length:751 Length:751
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## lbw lb lcb cb
## Length:751 Length:751 Length:751 Length:751

```



```
## Class :character   Class :character   Class :character   Class :character
## Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##      rcb            rb            gk            player_face_url
## Length:751         Length:751         Length:751         Length:751
## Class :character   Class :character   Class :character   Class :character
## Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
## club_logo_url      club_flag_url      nation_logo_url      nation_flag_url
## Length:751         Length:751         Length:751         Length:751
## Class :character   Class :character   Class :character   Class :character
## Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
```

Análisis

- El dataset contiene múltiples variables de rendimiento
- Existen columnas duplicadas debido al proceso de unión de datasets
- Se identifican variables numéricas y categóricas

3. Diagnóstico de Calidad de Datos

```
colSums(is.na(data))
```

```
##      sofifa_id_x      short_name_x
##           0           0
##      long_name      player_positions_x
##           0           0
##      overall_x      potential_x
##           0           0
##      age_x      nationality_name_x
##           0           0
##      nation_position_x      nation_jersey_number_x
##           382           382
##      club_name_x      league_name_x
##           10           10
##      value_eur_x      wage_eur_x
##           10           10
##      pace_x      shooting_x
```

##	77	77
##	passing_x	dribbling_x
##	77	77
##	defending_x	physic_x
##	77	77
##	sofifa_id_y	player_url
##	0	0
##	short_name_y	player_positions_y
##	0	0
##	overall_y	potential_y
##	0	0
##	value_eur_y	wage_eur_y
##	10	10
##	age_y	dob
##	0	0
##	height_cm	weight_kg
##	0	0
##	club_team_id	club_name_y
##	10	10
##	league_name_y	league_level
##	10	10
##	club_position	club_jersey_number
##	10	10
##	club_loaned_from	club_joined
##	719	42
##	club_contract_valid_until	nationality_id
##	10	0
##	nationality_name_y	nation_team_id
##	0	383
##	nation_position_y	nation_jersey_number_y
##	383	383
##	preferred_foot	weak_foot
##	0	0
##	skill_moves	international_reputation
##	0	0
##	work_rate	body_type
##	0	0
##	real_face	release_clause_eur
##	0	42
##	player_tags	player_traits
##	540	115
##	pace_y	shooting_y
##	76	76
##	passing_y	dribbling_y
##	76	76
##	defending_y	physic_y
##	76	76
##	attacking_crossing	attacking_finishing
##	0	0
##	attacking_heading_accuracy	attacking_short_passing
##	0	0
##	attacking_volleys	skill_dribbling
##	0	0
##	skill_curve	skill_fk_accuracy

##	0	0
##	skill_long_passing	skill_ball_control
##	0	0
##	movement_acceleration	movement_sprint_speed
##	0	0
##	movement_agility	movement_reactions
##	0	0
##	movement_balance	power_shot_power
##	0	0
##	power_jumping	power_stamina
##	0	0
##	power_strength	power_long_shots
##	0	0
##	mentality_aggression	mentality_interceptions
##	0	0
##	mentality_positioning	mentality_vision
##	0	0
##	mentality_penalties	mentality_composure
##	0	0
##	defending_marking_awareness	defending_standing_tackle
##	0	0
##	defending_sliding_tackle	goalkeeping_diving
##	0	0
##	goalkeeping_handling	goalkeeping_kicking
##	0	0
##	goalkeeping_positioning	goalkeeping_reflexes
##	0	0
##	goalkeeping_speed	ls
##	675	0
##	st	rs
##	0	0
##	lw	lf
##	0	0
##	cf	rf
##	0	0
##	rw	lam
##	0	0
##	cam	ram
##	0	0
##	lm	lcm
##	0	0
##	cm	rcm
##	0	0
##	rm	lwb
##	0	0
##	ldm	cdm
##	0	0
##	rdm	rwf
##	0	0
##	lb	lcb
##	0	0
##	cb	rcb
##	0	0
##	rb	gk

```
##              0              0
##      player_face_url      club_logo_url
##              0              10
##      club_flag_url      nation_logo_url
##              10             383
##      nation_flag_url
##              0
```

Análisis

- Se observan valores faltantes en variables de rendimiento
- Existen duplicaciones de columnas (`_x` y `_y`)
- Problemas derivados del proceso de merge

4. Limpieza y Preparación de Datos

```
data_clean <- data %>%
  # eliminar columnas duplicadas del merge
  select(-ends_with("_x")) %>%

  # renombrar columnas principales
  rename(
    overall = overall_y,
    pace = pace_y,
    shooting = shooting_y,
    passing = passing_y,
    dribbling = dribbling_y,
    defending = defending_y,
    physic = physic_y,
    age = age_y,
    nationality = nationality_name_y
  ) %>%

  # seleccionar variables relevantes
  select(
    long_name, overall, pace, shooting, passing,
    dribbling, defending, physic, age, nationality
  ) %>%

  # eliminar nulos críticos
  drop_na(overall)
```

Decisiones

- Se eliminaron columnas duplicadas (`_x`)
- Se priorizaron datos originales de SOFIFA (`_y`)
- Se seleccionaron variables relevantes para análisis

- Se eliminaron registros con valores faltantes críticos

5. Validación de Datos

```
summary(data_clean)
```

```
##   long_name          overall          pace          shooting
## Length:751      Min.    :62.00  Min.    :32.00  Min.    :22.00
## Class :character 1st Qu.:72.00  1st Qu.:66.00  1st Qu.:54.00
## Mode  :character Median :76.00  Median :74.00  Median :67.00
##              Mean  :76.26  Mean  :73.47  Mean  :62.85
##              3rd Qu.:80.00  3rd Qu.:81.00  3rd Qu.:73.00
##              Max.   :93.00  Max.   :97.00  Max.   :94.00
##              NA's    :76    NA's    :76    NA's    :76
##   passing      dribbling      defending      physic
## Min.    :38.00  Min.    :40.00  Min.    :20.00  Min.    :40.00
## 1st Qu.:62.00  1st Qu.:68.00  1st Qu.:43.00  1st Qu.:67.00
## Median :68.00  Median :73.00  Median :65.00  Median :73.00
## Mean    :67.93  Mean    :72.59  Mean    :59.07  Mean    :71.61
## 3rd Qu.:75.00  3rd Qu.:79.00  3rd Qu.:74.00  3rd Qu.:78.00
## Max.    :93.00  Max.    :95.00  Max.    :91.00  Max.    :89.00
## NA's    :76    NA's    :76    NA's    :76    NA's    :76
##   age      nationality
## Min.    :17.00  Length:751
## 1st Qu.:25.00  Class :character
## Median :27.00  Mode  :character
## Mean    :27.19
## 3rd Qu.:30.00
## Max.    :38.00
##
```

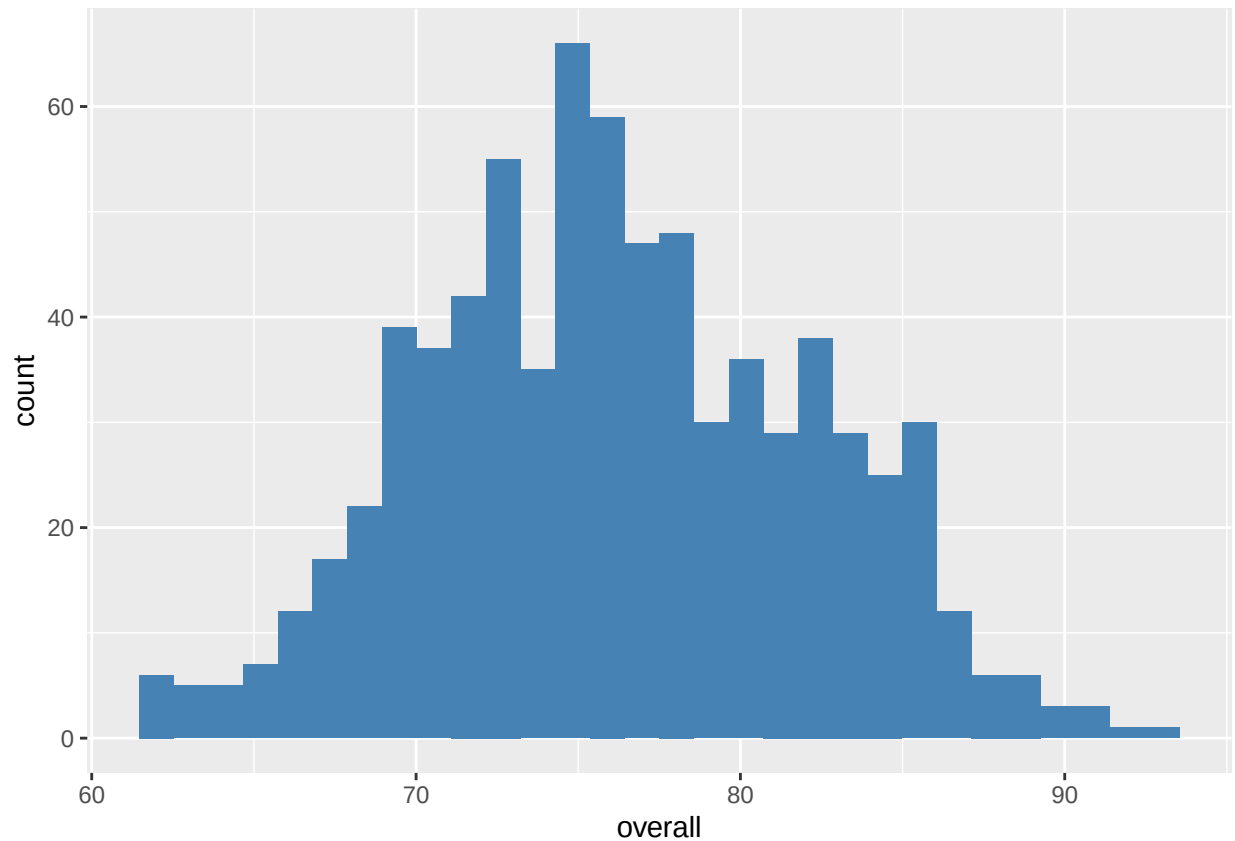
Análisis

- Los valores de **overall** están dentro de rangos esperados
- No se observan inconsistencias críticas
- Los datos son coherentes para análisis

6. Análisis Exploratorio (EDA)

Distribución del Overall

```
ggplot(data_clean, aes(x = overall)) +
  geom_histogram(bins = 30, fill = "steelblue")
```



Interpretación

La mayoría de los jugadores se concentran en valores medios de rendimiento (65-80).

Top 10 jugadores

```
top_players <- data_clean %>%
  arrange(desc(overall)) %>%
  head(10)
```

top_players

A tibble: 10 x 10

##	long_name	overall	pace	shooting	passing	dribbling	defending	physic	age
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
##	1 Lionel André~	93	85	92	91	95	34	65	34
##	2 Robert Lewan~	92	78	92	79	86	44	82	32
##	3 Kylian Mbapp~	91	97	88	80	92	36	77	22
##	4 Kevin De Bru~	91	76	86	93	88	64	78	30
##	5 Cristiano Ro~	91	87	94	80	88	34	75	36

```
## 6 Harry Kane          90    70    91    83    83    47    83    27
## 7 N’Golo Kanté        90    78    66    75    82    87    83    30
## 8 Manuel Peter~       90    NA    NA    NA    NA    NA    NA    35
## 9 Sadio Mané          89    91    83    80    89    44    77    29
## 10 Virgil van D~      89    78    60    71    72    91    84    29
## # i 1 more variable: nationality <chr>
```

Outliers (IQR)

```
Q1 <- quantile(data_clean$overall, 0.25)
Q3 <- quantile(data_clean$overall, 0.75)
IQR <- Q3 - Q1

outliers <- data_clean %>%
  filter(overall < Q1 - 1.5*IQR | overall > Q3 + 1.5*IQR)

nrow(outliers)
```

```
## [1] 1
```

7. Análisis Bivariado

Correlación

```
cor_matrix <- data_clean %>%
  select(overall, pace, shooting, passing, dribbling) %>%
  cor(use = "complete.obs")

cor_matrix
```

```
##          overall      pace shooting passing dribbling
## overall    1.0000000 0.1526711 0.4229248 0.6389075 0.5844586
## pace        0.1526711 1.0000000 0.3251928 0.1869815 0.4694388
## shooting    0.4229248 0.3251928 1.0000000 0.6629921 0.7715496
## passing     0.6389075 0.1869815 0.6629921 1.0000000 0.8340242
## dribbling   0.5844586 0.4694388 0.7715496 0.8340242 1.0000000
```

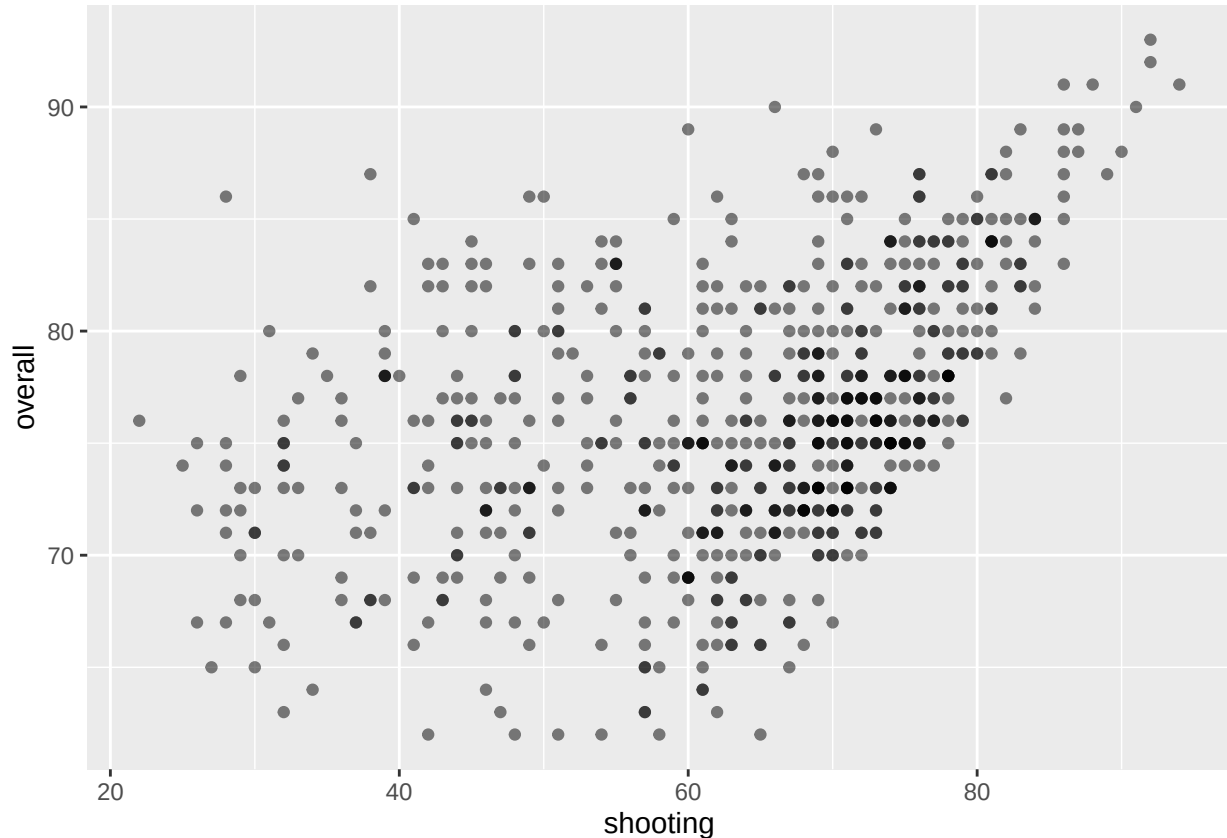
Interpretación

Las variables ofensivas presentan alta correlación con el overall, indicando su impacto en el rendimiento.

Relación Shooting vs Overall

```
ggplot(data_clean, aes(x = shooting, y = overall)) +  
  geom_point(alpha = 0.5)
```

```
## Warning: Removed 76 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



8. Transformación de Datos (KDD)

```
data_transformed <- data_clean %>%  
  mutate(  
    performance_level = case_when(  
      overall >= 85 ~ "Elite",  
      overall >= 75 ~ "Alto",  
      TRUE ~ "Medio"  
    )  
  )
```


Análisis

Se creó una variable categórica que permite clasificar a los jugadores según su rendimiento.

9. Análisis Final

Hallazgos

- Los jugadores con mayor shooting y passing tienen mejor overall
 - La mayoría de jugadores presentan rendimiento medio
 - Existen pocos jugadores de élite
-

10. Conclusiones

- El dataset permitió identificar patrones claros de rendimiento
 - La limpieza de datos fue clave para obtener resultados confiables
 - Las variables ofensivas tienen mayor impacto en el desempeño
-

11. Referencias

- SOFIFA dataset
- Metodología KDD